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package and methods

1. random

```
In [2]: import random
```

```
In [3]: # randint()
dir(random)
random.randint
random.randint(1,10)
```

```
Out[3]: 4
```

```
In [4]: # random()
random.random()
```

```
Out[4]: 0.20490762932863338
```

```
In [11]: # choice
random.choice(["Red", "Blue", "Green", "Yellow"])
```

```
Out[11]: 'Green'
```

```
In [13]: # sample
numbers=[10,20,30,40,50]
sample=random.sample(numbers,3)
sample
```

```
Out[13]: [40, 30, 10]
```

```
In [14]: # seed
random.seed(5)
random.randint(1,19)
```

```
Out[14]: 9
```

2.Math

- sqrt
- pow
- pi
- e

```
In [15]: import math
```

```
In [17]: # sqrt  
math.sqrt(10)
```

Out[17]: 3.1622776601683795

```
In [18]: # pow  
math.pow(1,10)
```

Out[18]: 1.0

```
In [19]: # pi  
math.pi
```

Out[19]: 3.141592653589793

```
In [20]: math.e
```

Out[20]: 2.718281828459045

3.time

- sleep
- time

```
In [21]: import time
```

```
In [22]: # sleep  
time.sleep(5)  
print('Nareshit')
```

Nareshit

```
In [23]: # time  
start=time.time()  
time.sleep(3)  
end=time.time()  
end-start
```

Out[23]: 3.0027923583984375

```
In [24]: time.time()
```

Out[24]: 1781676962.2090418

4.keywords

- kwlist

```
In [26]: import keyword
```

```
In [27]: dir()
```

```
Out[27]: ['In',
          'Out',
          '_',
          '_11',
          '_12',
          '_13',
          '_14',
          '_17',
          '_18',
          '_19',
          '_20',
          '_23',
          '_24',
          '_3',
          '_4',
          '_9',
          '_',
          '_',
          '_',
          '__builtin__',
          '__builtins__',
          '__doc__',
          '__loader__',
          '__name__',
          '__package__',
          '__session__',
          '__spec__',
          '_dh',
          '_i',
          '_i1',
          '_i10',
          '_i11',
          '_i12',
          '_i13',
          '_i14',
          '_i15',
          '_i16',
          '_i17',
          '_i18',
          '_i19',
          '_i2',
          '_i20',
          '_i21',
          '_i22',
          '_i23',
          '_i24',
          '_i25',
          '_i26',
          '_i27',
          '_i3',
          '_i4',
          '_i5',
          '_i6',
          '_i7',
          '_i8',
          '_i9',
          '_ih',
          '_ii',
          '_iii',
          '_oh',
          'end',
```

```
'exit',  
'get_ipython',  
'keyword',  
'math',  
'numbers',  
'open',  
'quit',  
'random',  
'sample',  
'start',  
'time']
```

```
In [29]: import keyword  
keyword.kwlist
```

```
Out[29]: ['False',  
'None',  
'True',  
'and',  
'as',  
'assert',  
'async',  
'await',  
'break',  
'class',  
'continue',  
'def',  
'del',  
'elif',  
'else',  
'except',  
'finally',  
'for',  
'from',  
'global',  
'if',  
'import',  
'in',  
'is',  
'lambda',  
'nonlocal',  
'not',  
'or',  
'pass',  
'raise',  
'return',  
'try',  
'while',  
'with',  
'yield']
```

5.os

- getcwd
- makedirs

```
In [30]: import os
```

```
In [31]: os.getcwd()
```

```
Out[31]: 'C:\\Users\\LENOVO\\Documents\\NareshIT\\Python_Sessions'
```

```
In [35]: dir(os)
```

```
Out[35]: ['DirEntry',
          'EX_OK',
          'F_OK',
          'GenericAlias',
          'Mapping',
          'MutableMapping',
          'O_APPEND',
          'O_BINARY',
          'O_CREAT',
          'O_EXCL',
          'O_NOINHERIT',
          'O_RANDOM',
          'O_RDONLY',
          'O_RDWR',
          'O_SEQUENTIAL',
          'O_SHORT_LIVED',
          'O_TEMPORARY',
          'O_TEXT',
          'O_TRUNC',
          'O_WRONLY',
          'P_DETACH',
          'P_NOWAIT',
          'P_NOWAITO',
          'P_OVERLAY',
          'P_WAIT',
          'PathLike',
          'R_OK',
          'SEEK_CUR',
          'SEEK_END',
          'SEEK_SET',
          'TMP_MAX',
          'W_OK',
          'X_OK',
          '_AddedDllDirectory',
          '_Environ',
          '__all__',
          '__builtins__',
          '__cached__',
          '__doc__',
          '__file__',
          '__loader__',
          '__name__',
          '__package__',
          '__spec__',
          '_check_methods',
          '_execvpe',
          '_exists',
          '_exit',
          '_fspath',
          '_get_exports_list',
          '_walk_symlinks_as_files',
          '_wrap_close',
          'abc',
          'abort',
          'access',
          'add_dll_directory',
          'altsep',
          'chdir',
          'chmod',
          'close',
```

```
'closerange',  
'cpu_count',  
'curdir',  
'defpath',  
'device_encoding',  
'devnull',  
'dup',  
'dup2',  
'environ',  
'error',  
'execl',  
'execle',  
'execlp',  
'execlpe',  
'execv',  
'execve',  
'execvp',  
'execvpe',  
'extsep',  
'fchmod',  
'fdopen',  
'fsdecode',  
'fsencode',  
'fspath',  
'fstat',  
'fsync',  
'ftruncate',  
'get_blocking',  
'get_exec_path',  
'get_handle_inheritable',  
'get_inheritable',  
'get_terminal_size',  
'getcwd',  
'getcwdb',  
'getenv',  
'getlogin',  
'getpid',  
'getppid',  
'isatty',  
'kill',  
'lchmod',  
'linesep',  
'link',  
'listdir',  
'listdrives',  
'listmounts',  
'listvolumes',  
'lseek',  
'lstat',  
'makedirs',  
'mkdir',  
'name',  
'open',  
'pardir',  
'path',  
'pathsep',  
'pipe',  
'popen',  
'process_cpu_count',  
'putenv',
```

```

'read',
'readlink',
'remove',
'removedirs',
'rename',
'renames',
'replace',
'rmdir',
'scandir',
'sep',
'set_blocking',
'set_handle_inheritable',
'set_inheritable',
'spawnl',
'spawnle',
'spawnv',
'spawnve',
'st',
'startfile',
'stat',
'stat_result',
'statvfs_result',
'strerror',
'supports_bytes_environ',
'supports_dir_fd',
'supports_effective_ids',
'supports_fd',
'supports_follow_symlinks',
'symlink',
'sys',
'system',
'terminal_size',
'times',
'times_result',
'truncate',
'umask',
'uname_result',
'unlink',
'unsetenv',
'urandom',
'utime',
'waitpid',
'waitstatus_to_exitcode',
'walk',
'write']

```

```

In [45]: os.makedirs("naresh12")
         print("folder created successfully")

```

folder created successfully

6.Date

- today date

```

In [49]: from datetime import date

         today = date.today()

```



```
print("Today's date is:", today)
```

Today's date is: 2026-06-17

7. numpy

- min
- max
- sum
- square

```
In [57]: import numpy
```

```
In [60]: import numpy as np  
a=[10,12,14,15]  
print("minimum value:",np.min(a))
```

minimum value: 10

```
In [61]: print("maximum value :",np.max(a))
```

maximum value : 15

```
In [62]: print("sum of values :",np.sum(a))
```

sum of values : 51

```
In [63]: print("square of values :",np.square(a))
```

square of values : [100 144 196 225]

```
In [ ]:
```